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Spring-Data-JPA-Hibernate

Difference Between JPA, Hibernate, and Spring Data JPA

Java Persistence API (JPA)

- JPA is a Java specification (JSR 338) for persisting, reading, and managing data from Java objects.
- It does not contain any concrete implementation.
- It is a set of interfaces and annotations.

Hibernate

- Hibernate is an ORM (Object-Relational Mapping) tool that implements the JPA specification.
- It provides concrete implementation of all JPA interfaces.
- It manages database operations and SQL generation.

Spring Data JPA

- Spring Data JPA does not have its own JPA implementation.
- It reduces boilerplate code required to implement data access layers.
- It is another level of abstraction over JPA implementation providers like Hibernate.
- It automatically manages transactions.

Exercise 2: Spring Data JPA - Quick Example

Objective

Create a Spring Boot application using Spring Data JPA and Hibernate to manage country data in a MySQL database.

Steps

1. A Spring Boot project is created using start.spring.io with the following configurations:
 - Group: com.cognizant
 - Artifact: orm-learn
 - Description: "Demo project for Spring Data JPA and Hibernate"
 - Dependencies: Spring Boot DevTools, Spring Data JPA, MySQL Driver
2. A new schema ormllearn is created in MySQL database.
3. The application.properties file is configured with database connection details and logging configuration.
4. The country table is created in MySQL with columns for code and name, and sample records are inserted.
5. A package com.cognizant.ormlearn.model is created with Country.java entity class using @Entity, @Table, @Id, and @Column annotations.
6. A package com.cognizant.ormlearn.repository is created with CountryRepository interface extending JpaRepository<Country, String> and @Repository annotation.
7. A package com.cognizant.ormlearn.service is created with CountryService class using @Service annotation, autowired CountryRepository, and a getAllCountries() method with @Transactional.
8. The OrmLearnApplication main class is modified to load the Spring context, get the CountryService bean, and call testGetAllCountries() method.
9. The application is executed and verified that country data is retrieved from the database.

Key Concepts

- @Entity marks a class as a JPA entity that maps to a database table
- @Table(name="country") specifies the table name in the database
- @Id marks the primary key field
- @Column(name="co_code") maps a field to a specific database column
- JpaRepository<Country, String> provides built-in CRUD operations
- @Repository marks an interface as a Spring Data repository
- @Service marks a class as a service layer component
- @Transactional ensures the method runs within a transaction boundary
- @Autowired injects the repository dependency automatically
- Spring Data JPA automatically implements the repository interface at runtime

The application displays the list of countries retrieved from the MySQL database:

```
countries=[Country [code=IN, name=India], Country [code=US, name=United States of America]]
```

Output:

```

  02-07-26 23:20:06.899 restartedMain INFO c.c.o.OmLearnApplication logStarting 55 Starting OmLearnApplication using Java 26.0.1 on ADARSHs-MacBook-Air-2.1
  02-07-26 23:20:06.899 restartedMain INFO c.c.o.OmLearnApplication logStartupProfileInfo 632 No active profile set, falling back to 1 default profile: "default"
  02-07-26 23:20:06.900 restartedMain INFO ertyDefaultsPostProcessor logTo 255 Devtools property defaults active! Set 'spring.devtools.add-properties' t
  02-07-26 23:20:06.931 restartedMain o 'false' to disable
  02-07-26 23:20:07.163 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 132 Bootstrapping Spring Data JPA repositories in DEFAULT mode.
  02-07-26 23:20:07.189 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 201 Finished Spring Data repository scanning in 21 ms. Found 1 JPA repository
  02-07-26 23:20:07.425 restartedMain INFO o.h.j.i.util.LogHelper logPersistenceUnitInformation 31 HHN000204: Processing PersistenceUnitInfo [name: default]
  02-07-26 23:20:07.452 restartedMain INFO org.hibernate.Version logVersion 44 HHN000412: Hibernate ORM core version 5.6.15.Final
  02-07-26 23:20:07.544 restartedMain INFO o.h.a.common.Version <clinit> 56 HCANN000001: Hibernate Commons Annotations {5.1.2.Final}
  02-07-26 23:20:07.603 restartedMain INFO c.z.h.HikariDataSource getConnection 110 HikariPool-1 - Starting...
  02-07-26 23:20:07.901 restartedMain INFO c.z.h.HikariDataSource getConnection 123 HikariPool-1 - Start completed.
  02-07-26 23:20:07.919 restartedMain INFO o.h.dialect.Dialect <init> 175 HHN000400: Using dialect: org.hibernate.dialect.MySQL5Dialect
  02-07-26 23:20:08.222 restartedMain INFO .p.i.JtaPlatformInitiator initiateService 52 HHN000490: Using JtaPlatform implementation: [org.hibernate.engine.transa
  02-07-26 23:20:08.227 restartedMain ction.JtaPlatform.Internal.NoJtaPlatform]
  02-07-26 23:20:08.227 restartedMain INFO r.FntityManagerFactoryBean buildNativeEntityEntityManagerFactory 437 Initialized Jpa EntityManagerFactory for persistence unit 'default'
  02-07-26 23:20:08.423 restartedMain INFO OptionalLiveReloadServer startServer 59 LiveReload server is running on port 35729
  02-07-26 23:20:08.434 restartedMain INFO c.c.o.OmLearnApplication logStarted 61 Started OmLearnApplication in 1.756 seconds (JVM running for 2.024)
  02-07-26 23:20:08.436 restartedMain INFO c.c.o.OmLearnApplication main 22 Inside main
  02-07-26 23:20:08.436 restartedMain INFO c.c.o.OmLearnApplication testGetAllCountries 27 Start
  02-07-26 23:20:08.508 restartedMain DEBUG org.hibernate.SQL logStatement 144 select country0_co_code as co_code1_0, country0_co_name as co_name2_0_
  02-07-26 23:20:08.513 restartedMain TRACE o.h.t.d.s.BasicExtractor extract 60 extracted value [{co_code1_0} : [VARCHAR]] - [IN]
  02-07-26 23:20:08.516 restartedMain TRACE o.h.t.d.s.BasicExtractor extract 60 extracted value [{co_name2_0} : [VARCHAR]] - [India]
  02-07-26 23:20:08.516 restartedMain TRACE o.h.t.d.s.BasicExtractor extract 60 extracted value [{co_name2_0} : [VARCHAR]] - [US]
  02-07-26 23:20:08.517 restartedMain TRACE o.h.t.d.s.BasicExtractor extract 60 extracted value [{co_name2_0} : [VARCHAR]] - [United States of America]
  02-07-26 23:20:08.526 restartedMain DEBUG c.c.o.OmLearnApplication testGetAllCountries 29 countries=[Country {code=IN, name=India}, Country {code=US, name=United S
  02-07-26 23:20:08.526 restartedMain INFO c.c.o.OmLearnApplication testGetAllCountries 30 End
  02-07-26 23:20:08.531 licationShutdownHook INFO r.FntityManagerFactoryBean destroy 651 Closing Jpa EntityManagerFactory for persistence unit 'default'
  02-07-26 23:20:08.532 licationShutdownHook INFO c.z.h.HikariDataSource close 350 HikariPool-1 - Shutdown initiated...
  02-07-26 23:20:08.536 licationShutdownHook INFO c.z.h.HikariDataSource close 352 HikariPool-1 - Shutdown completed.

[INFO] BUILD SUCCESS
[INFO] Total time: 4.082 s
[INFO] Finished at: 2026-07-02T23:20:08+05:30
[INFO]

```